

Investigation of Long Short-Term Memorys to Temperature Prediction for Permanent Magnet Synchronous Motors

Oliver Wallachowitz, Wilhelm Kierkegaard, Joachim Böcker
Department of Mechatronics and Electrical Drive
Paderborn University
33098 Paderborn, Germany
Email: wallachowitz@uni-paderborn.de

Abstract—Heat being a major component in permanent magnet synchronous motors (PMSMs) is critical in many applications and temperature prediction is well known. The application of long short-term memory (LSTM) networks for temperature prediction is investigated. LSTM networks are a special type of recurrent neural network (RNN) which are able to learn long-term dependencies. In this paper, the LSTM network is used to predict the temperature of a PMSM. The LSTM network is trained on a dataset of temperature measurements and the resulting prediction is compared to the actual temperature. The results show that the LSTM network is able to predict the temperature of a PMSM with a high accuracy. The LSTM network is also able to predict the temperature of a PMSM over a long time period. The results show that the LSTM network is a good choice for temperature prediction in PMSMs.

1. INTRODUCTION

Permanent magnet synchronous motors (PMSMs) are preferred in many industrial applications (e.g. electrical vehicles) due to their prime design and power density as well as efficiency. To ensure safe operation in the long term and to extend the service life of the motor, it is necessary to predict the temperature of the motor. The temperature of the motor is a critical parameter for the motor's performance. For example, the motor's winding resistance increases with temperature, which leads to a higher copper loss. Therefore, it is important to predict the temperature of the motor. In this paper, the LSTM network is used to predict the temperature of a PMSM. The LSTM network is trained on a dataset of temperature measurements and the resulting prediction is compared to the actual temperature. The results show that the LSTM network is able to predict the temperature of a PMSM with a high accuracy. The LSTM network is also able to predict the temperature of a PMSM over a long time period. The results show that the LSTM network is a good choice for temperature prediction in PMSMs.

(2). In both cases, temperatures are determined by solving the thermal properties of electric motor parameters, e.g. the winding resistance or the total heat capacity. However, in the temperature-dependent parameter scenario, at very low and custom-tailored solutions can lead to significant prediction errors.

Direct thermal analysis can be avoided by using computer-aided design (CAD) and finite element analysis (FEA) methods [1]. However, FEA and CFD require enormous computational resources making them suitable for only high-end engineering simulation and in the medium run. A proper alternative to determine critical temperature components are lumped parameter thermal networks (LPTNs). These are based on the idea to simplify the complex thermal behavior of a given system and to determine the heat transfer processes by using equivalent circuit diagrams [2–4]. LPTNs can be gained at two ways. On the one hand, where LPTN approaches are based on geometrical and material component information leading to differential algebraic models of high order [5]. Second order control-oriented models are required to calculate them under real-time conditions which typically require industrial processing platforms. On the other hand, grey-box LPTNs incorporate empirical training data based on extensive measurements resulting in differentially reduced model orders but also increased system order [6]. Due to limited degrees of model order, grey-box LPTNs can be easily processed under real-time conditions – however, it is questionable if the grey-box LPTN approaches are the best choice for very detailed temperature estimation in high-powered systems, e.g. electrical drives, when it should be about the exact direct thermal model. However, extensive knowledge of the machine low characteristics leading to additional model-specific model errors for an engineering design.

This contribution presents grey-box reduced model networks (LPTNs) and presents suitable alternatives for the above applications. The task is to determine the additional degrees of freedom using LPTNs instead of abstract LPTNs trying to account the transient behavior beyond the state of the art. In literature, only few publications can be found in this field, e.g. using feedforward (FFNN) [7] or recurrent neural